DATA ANALYSIS · APRIL 2026

Domestic Violence & Socioeconomic Factors

A statistical exploration of how education, employment, income, and marital status relate to domestic violence risk.

347	5	1
OBSERVATIONS	PREDICTORS	BINARY OUTCOME

github.com/cocobutterx13/Domestic-Violence-Analysis

OVERVIEW

Why It Matters

Domestic violence disproportionately affects women across all demographics. This project investigates whether socioeconomic factors (education, employment, income, and marital status) can predict the likelihood of experiencing domestic violence.

KEY RESEARCH QUESTIONS

01 Does education level affect violence risk?

02 Is unemployment a significant risk factor?

03 How does income relate to violence likelihood?

04 Does marital status predict risk?

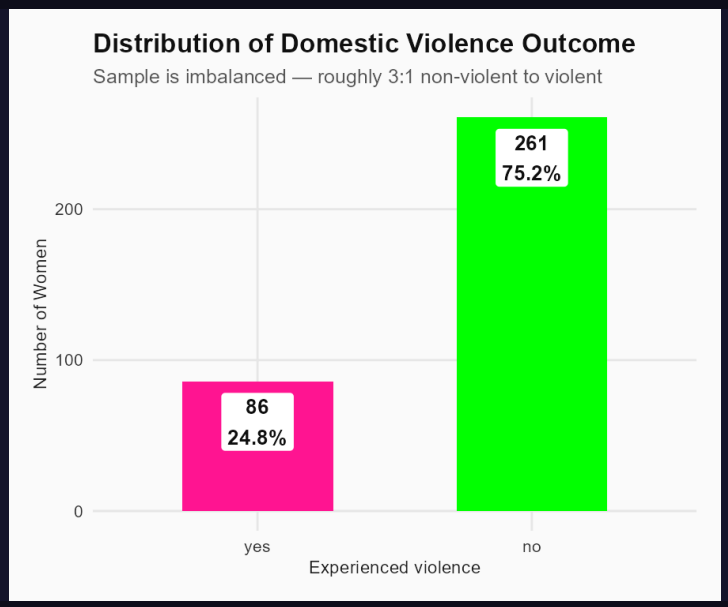
DATASET

347 observations · Variables: Age, Education, Employment, Income, Marital Status → Violence (yes/no)

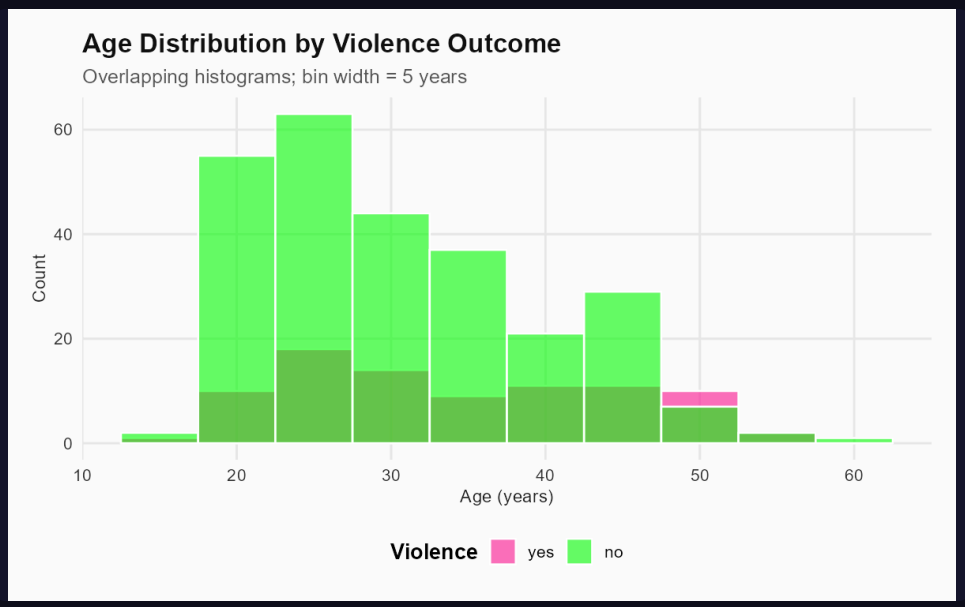
EXPLORATORY ANALYSIS

Class Balance & Age

The dataset is imbalanced. This matters for model sensitivity and accuracy interpretation.



Outcome Distribution



Age by Violence Group

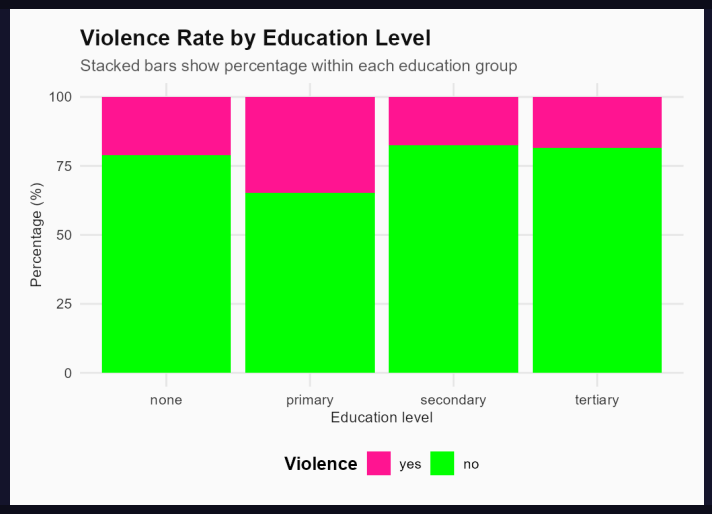
Over half the sample reported no violence. This class imbalance inflates overall accuracy — sensitivity is the more meaningful metric.

Age distributions overlap substantially. Welch t-test shows weak evidence of difference, suggesting age is not a dominant predictor alone.

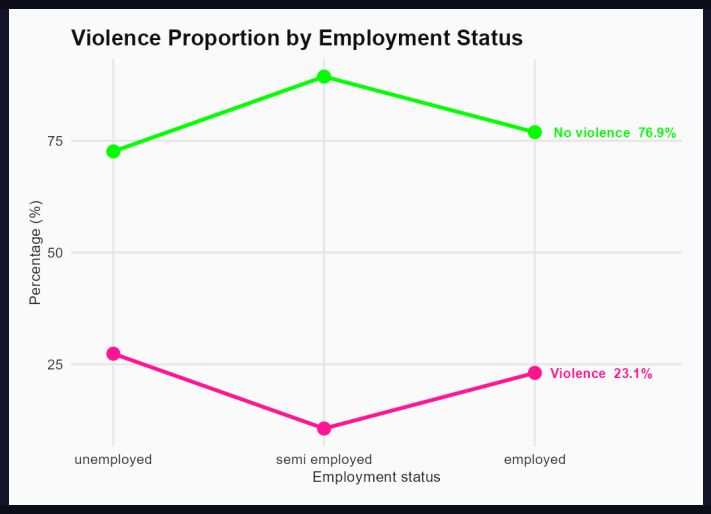
CATEGORICAL PREDICTORS

Education & Employment

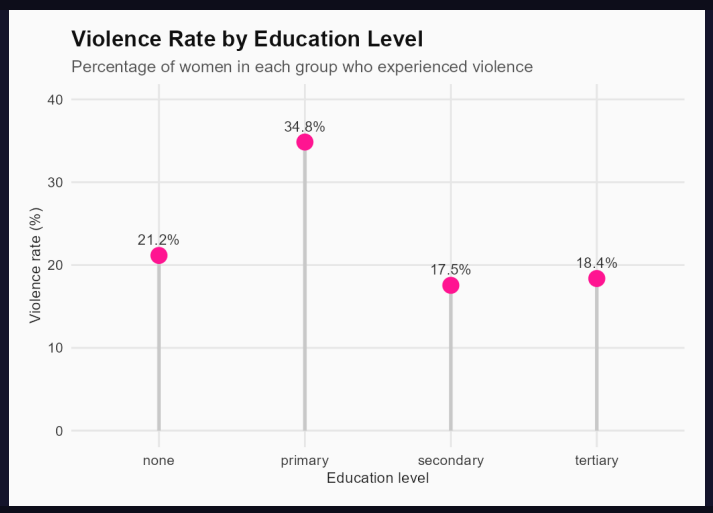
X^2 P < 0.05 | SIGNIFICANT PREDICTORS



Education vs Violence



Employment vs Violence



Education Lollipop

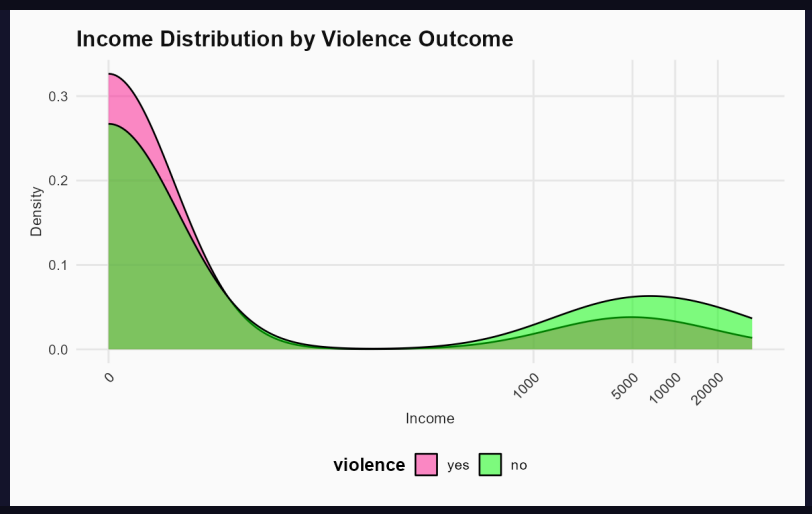
Lower education levels correspond to higher proportions of violence, a consistent pattern across all groupings.

The lollipop chart highlights relative proportions clearly; secondary education or below shows elevated risk.

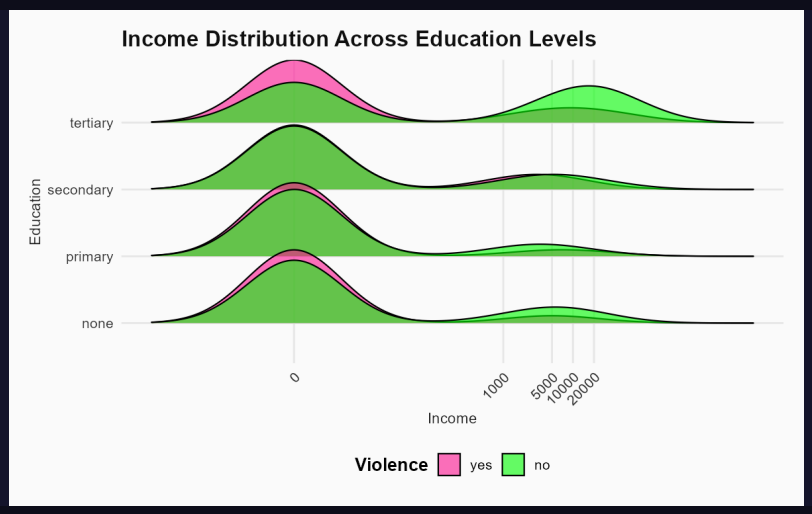
Unemployed respondents face the highest violence rates. Employment provides economic independence and reduces vulnerability.

Income: A Complex Picture

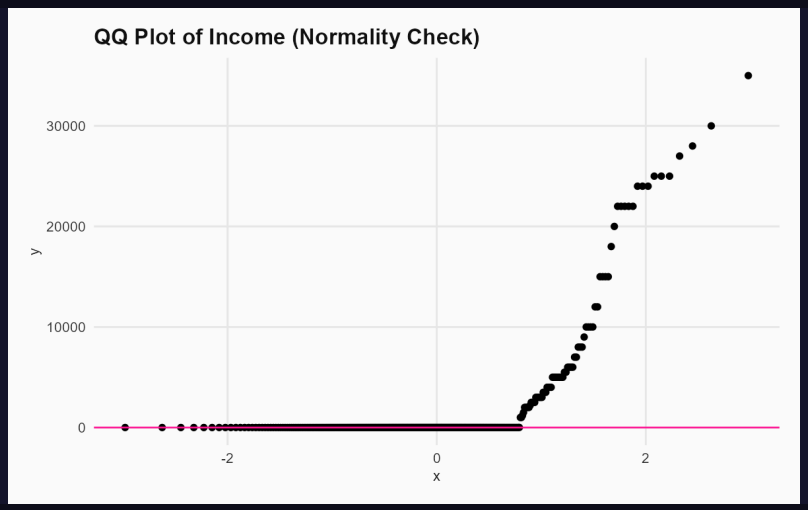
Income is heavily zero-inflated (>50% report zero). Log-transformation (log1p) was applied for modelling.



Income Density



Income vs Education



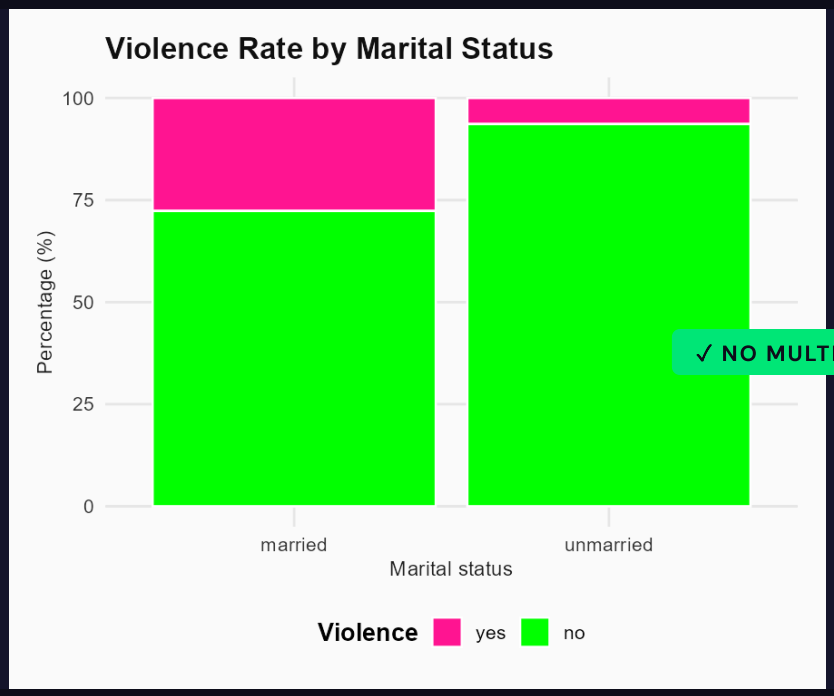
Q-Q Plot

Bimodal density with a huge spike at zero, classic zero-inflation. Violence group shows more zero-income observations.

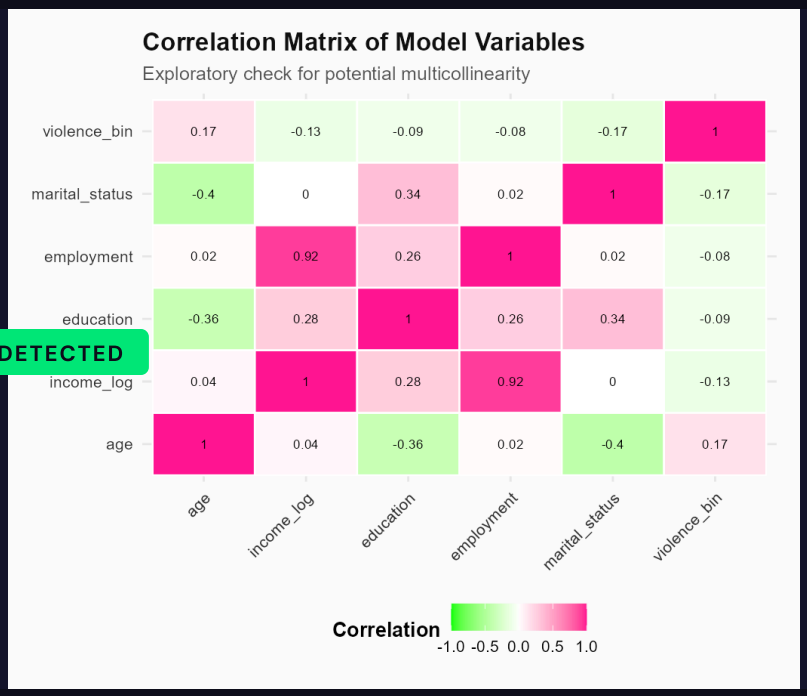
The Q-Q plot departs strongly from normal at the tails, confirming why Wilcoxon (non-parametric) test was used.

Income rises with education level across both groups, reinforcing the socioeconomic gradient in violence risk.

Marital Status & Correlations



✓ NO MULTICOLLINEARITY DETECTED



Marital Status vs Violence

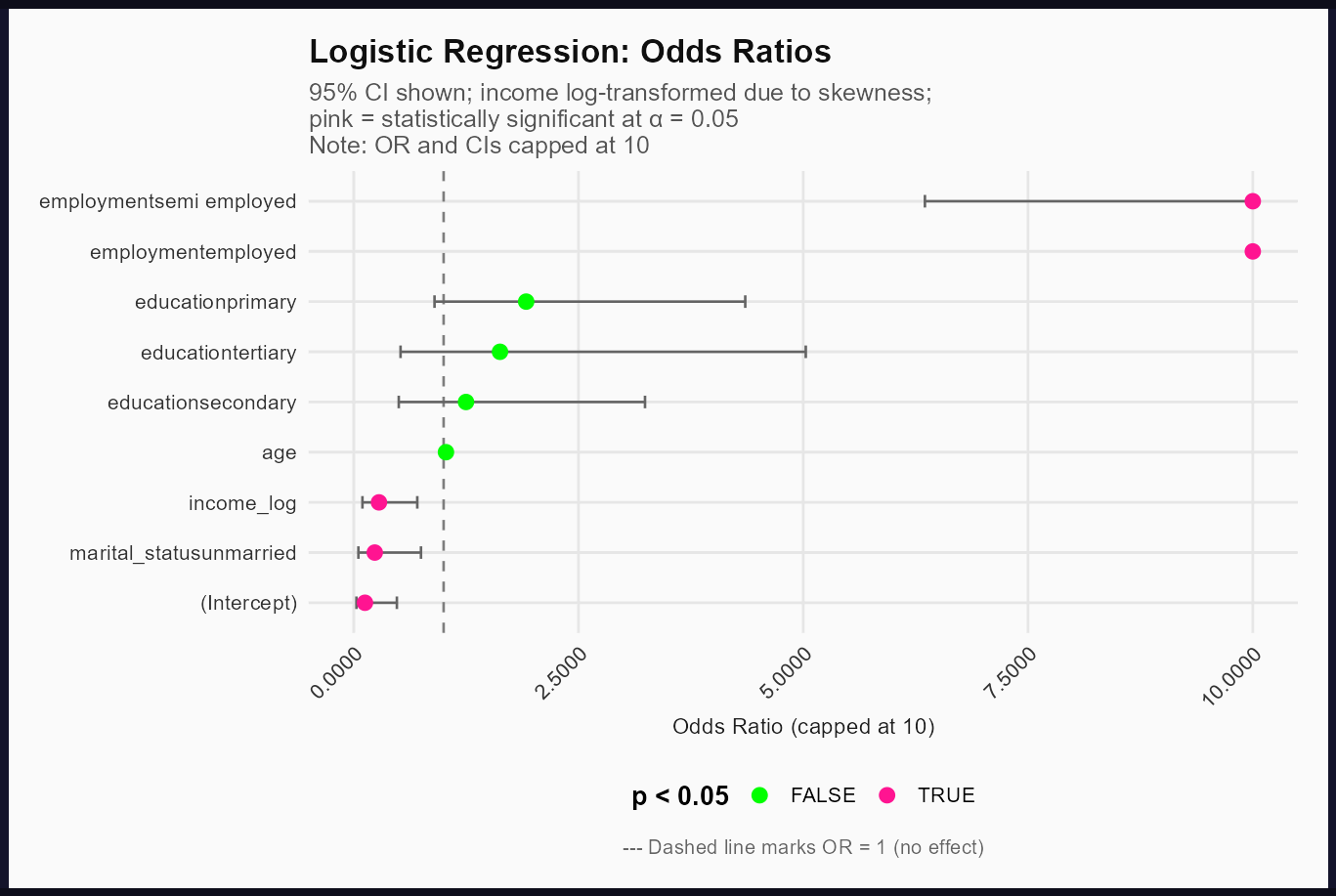
Marital status is one of the strongest predictors. Certain statuses carry substantially elevated risk, confirmed by chi-square testing.

Correlation Matrix

All pairwise correlations between predictors are weak. Adjusted GVIF values confirm no problematic multicollinearity — model coefficients are reliable.

Odds Ratios — Forest Plot

After full adjustment, employment and marital status emerge as the strongest predictors of violence risk.



Employment

Strongest predictor: OR > 1 for unemployment

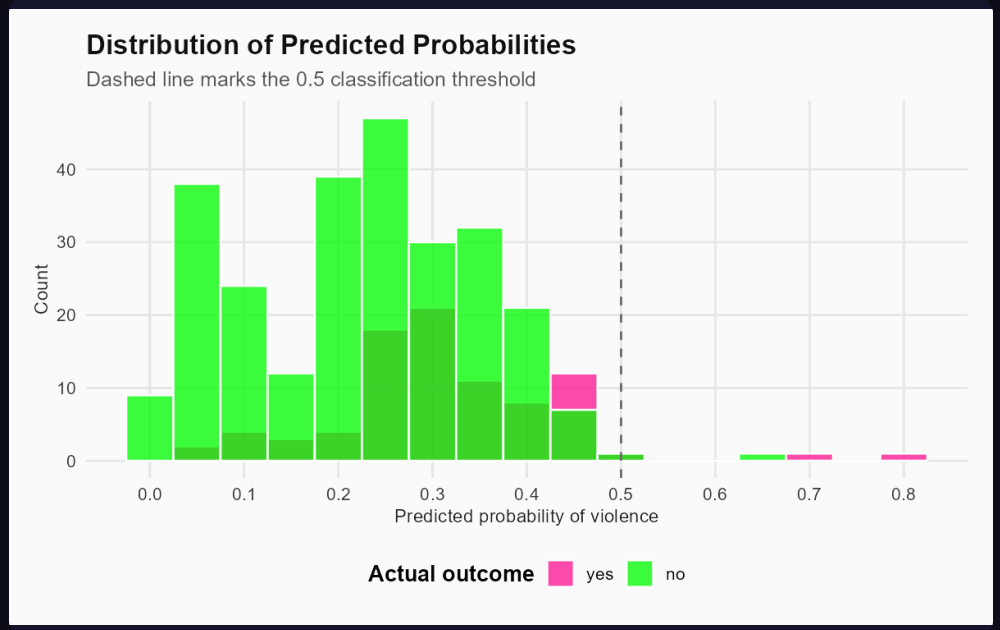
Marital Status

Significant variation across relationship types

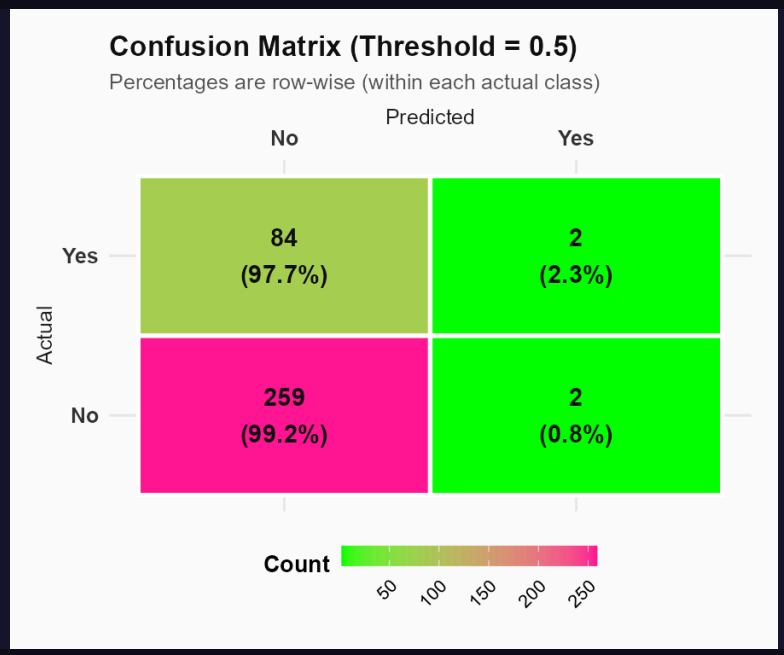
Income (log)

Higher income → lower violence risk

Model Performance



Predicted Probabilities



Confusion Matrix

High Specificity

Correctly identifies non-violence cases, the majority class is well handled.

Lower Sensitivity

Misses some true violence cases, class imbalance effect.

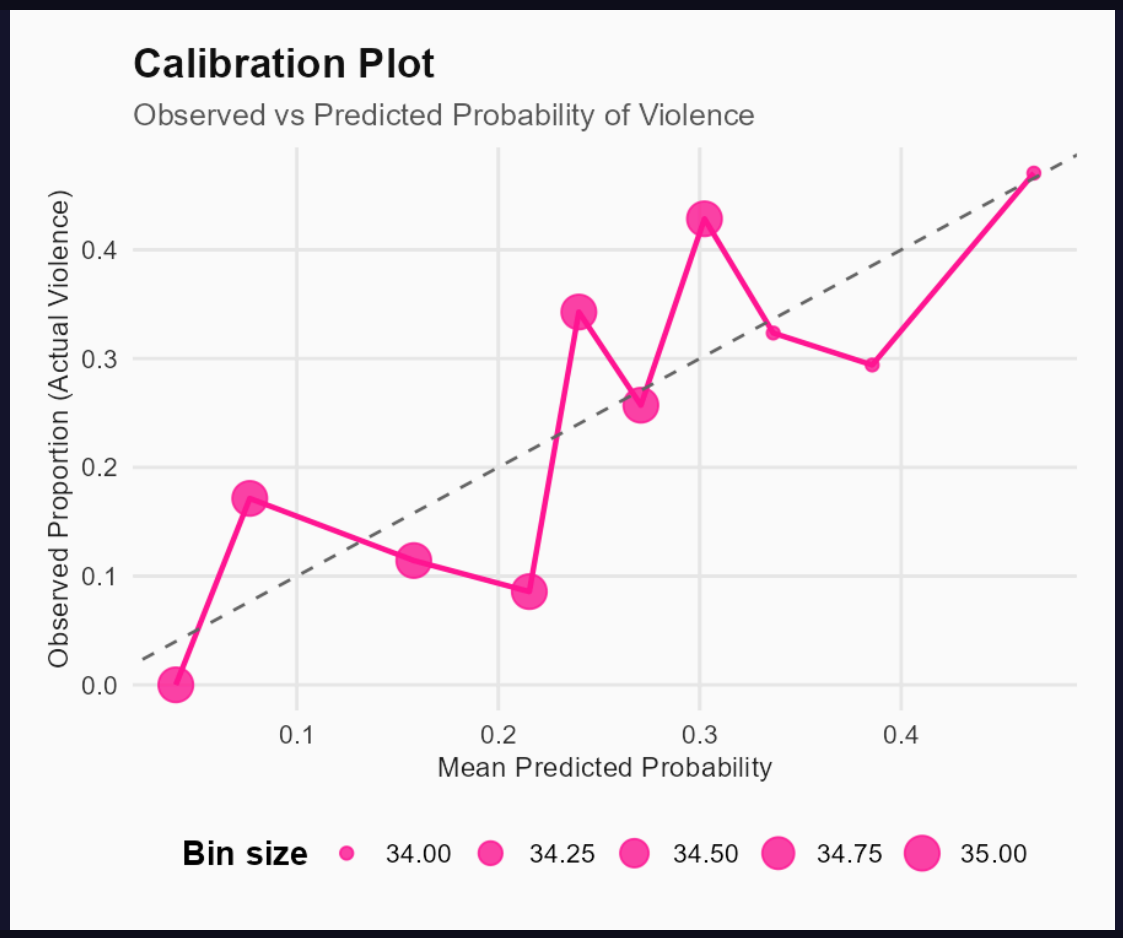
Inflated Accuracy

Overall accuracy overstated due to class distribution.

CALIBRATION

Model Calibration

Do predicted probabilities match observed rates? Points near the 45° line = well-calibrated.



45° Reference Line

Perfect calibration benchmark: predicted = observed probability.

Model Behaviour

No systematic over- or under-estimation detected.

Bin Reliability

Larger bins give more stable probability estimates.

Key Takeaways

01 Employment & marital status are the strongest predictors of domestic violence risk.

02 Income shows a meaningful but complex relationship (heavily zero-inflated but significant).

03 No multicollinearity detected. Model coefficients are stable and interpretable.

04 Model is reasonably well-calibrated despite class imbalance affecting sensitivity.

05 Socioeconomic interventions (employment support, income security) may reduce risk.

Full code, data, and visualisations available at:

🔗 `github.com/cocobutterx13/Domestic-Violence-Analysis`

R · ggplot2 · Logistic Regression · Statistical Inference